

Precalculus Review (Homework)

Simplify the following expressions. Do not leave your answers with negative exponents or fraction exponents.

$$\textcircled{1} x^{-\frac{1}{2}} \left(\frac{\sqrt[4]{x^9} a^2}{\left(\frac{1}{2}\right)^0 a b^2 \cdot b^2} \right)^3$$

$$\textcircled{2} \left(\frac{a}{b^3} \right)^{\frac{1}{2}} \left(\frac{x^4 a^{-1} x^2 b^2}{y^8 b^{-3}} \right)^{-2}$$

Factor the expressions below. Write the type of factoring used in the margin.

$$\textcircled{3} 8x^5 - 18x^4 - 100x^3 + 2x$$

$$\textcircled{4} b^2 - 49$$

$$\textcircled{5} x^2 + x - 2$$

$$\textcircled{6} 100x^2 - 81$$

$$\textcircled{7} x^2 - 7x + 10$$

$$\textcircled{8} 5x^8 + 20x^6 + 30x^3$$

$$\textcircled{9} 3x^2 + 8x + 4$$

$$\textcircled{10} 10x^2 - 3x - 18$$

Simplify.

$$\textcircled{11} \sqrt{10} \cdot \sqrt{7}$$

$$\textcircled{12} \sqrt[3]{2y} \cdot \sqrt[3]{y}$$

$$\textcircled{13} \sqrt[4]{\frac{81}{16}}$$

Factor out the perfect square/cube.

$$\textcircled{14} \sqrt{80}$$

$$\textcircled{15} \sqrt[3]{162}$$

Rationalize the denominators.

$$\textcircled{16} \frac{7}{\sqrt{5}}$$

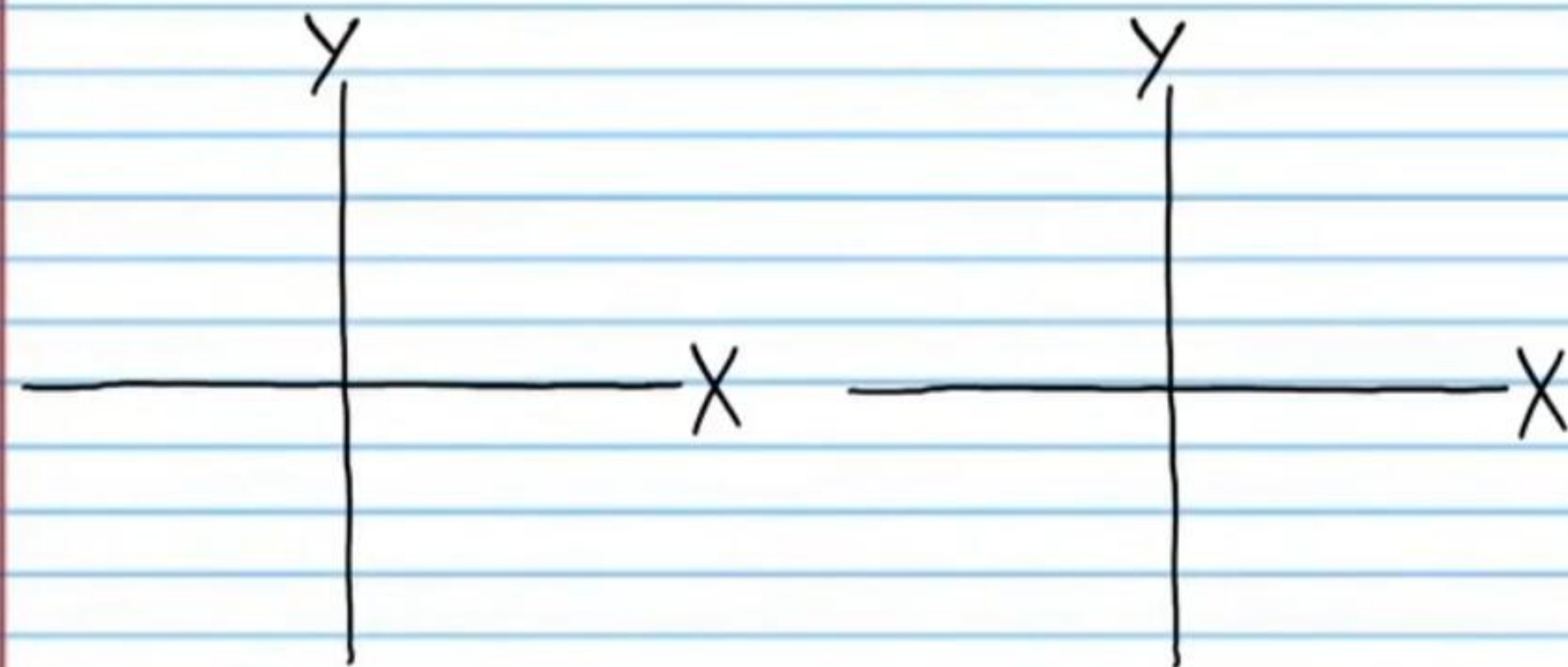
$$\textcircled{17} \frac{5}{\sqrt{40}}$$

$$\textcircled{18} \sqrt{\frac{45}{13}}$$

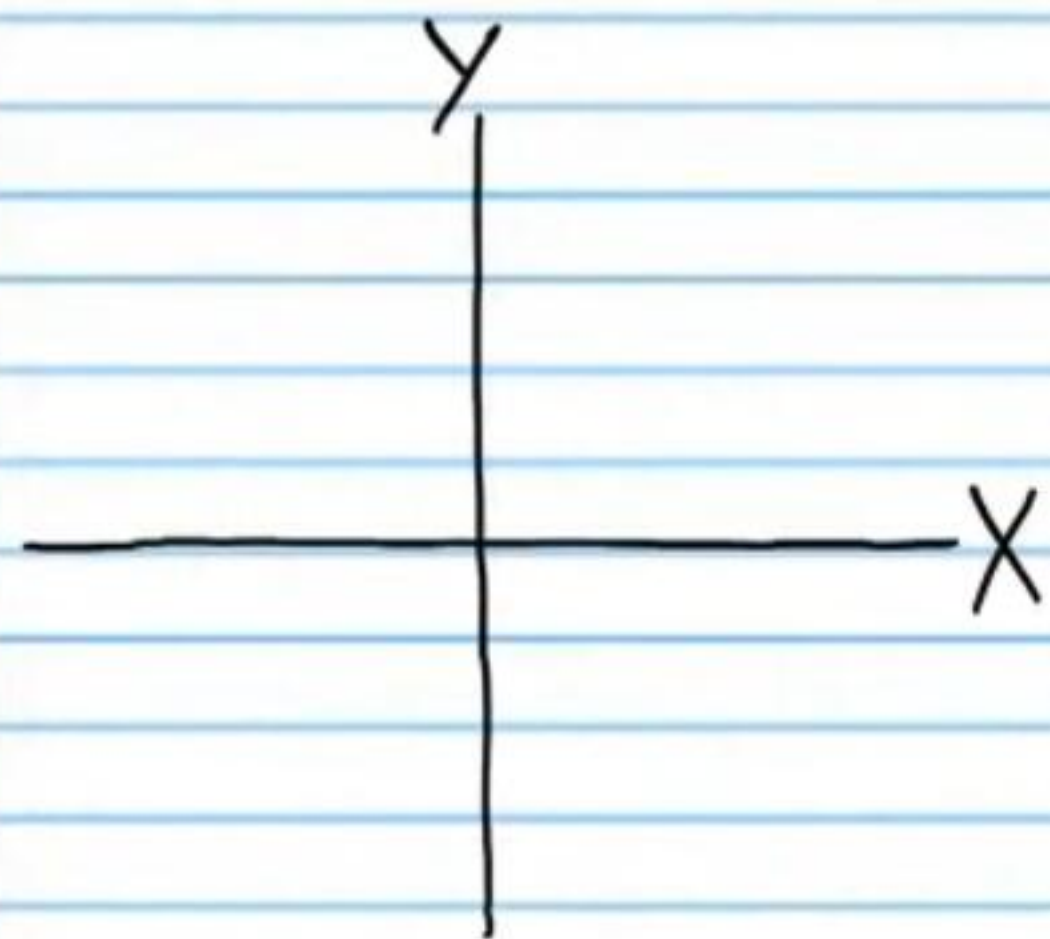
Find the value of each expression below.

$$\textcircled{19} \sin \frac{\pi}{4}$$

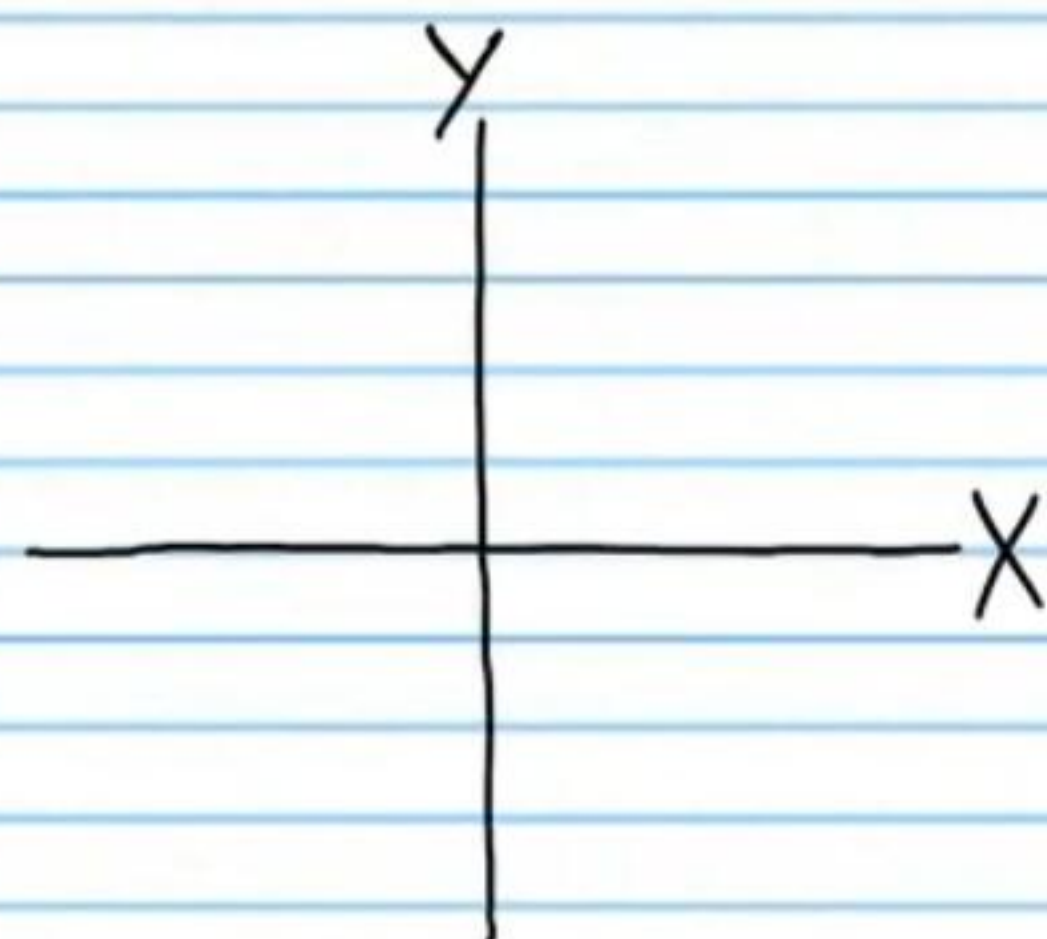
$$\textcircled{20} \sec \left(\frac{2\pi}{3} \right)$$



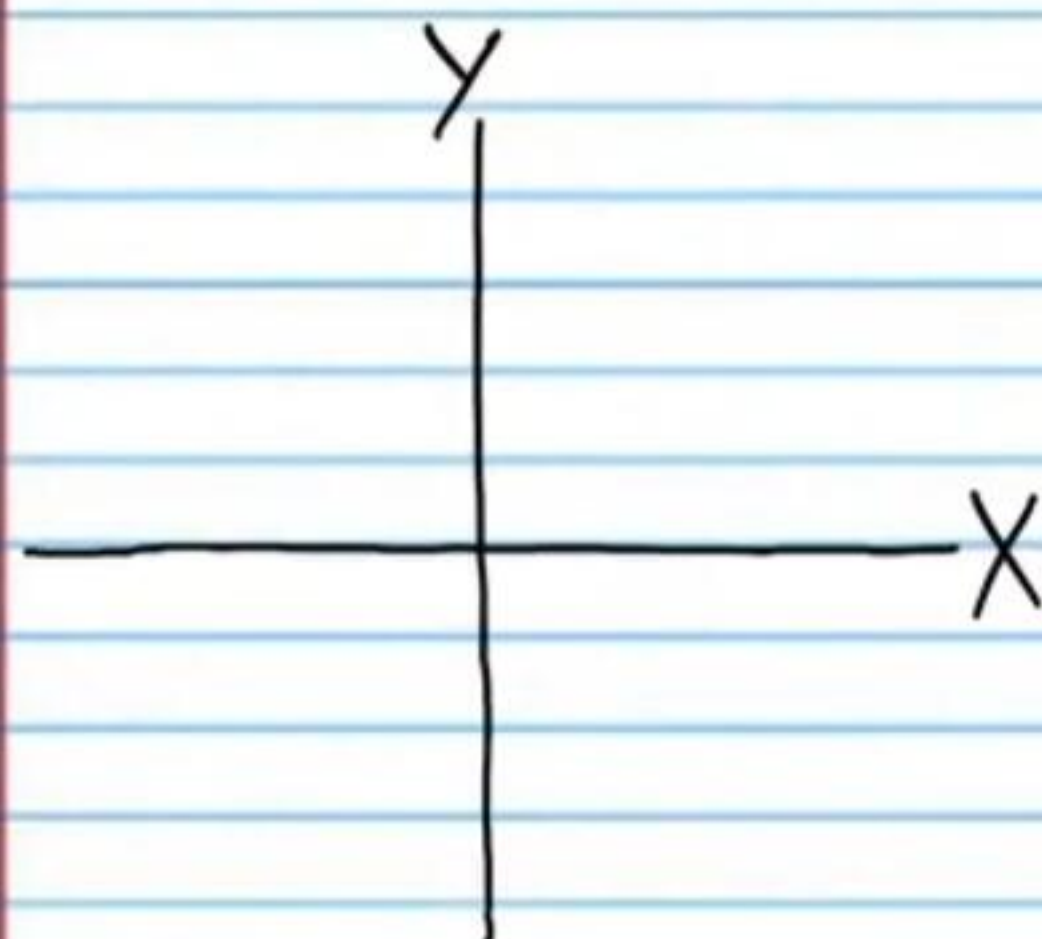
21) $\tan\left(\frac{7\pi}{6}\right)$



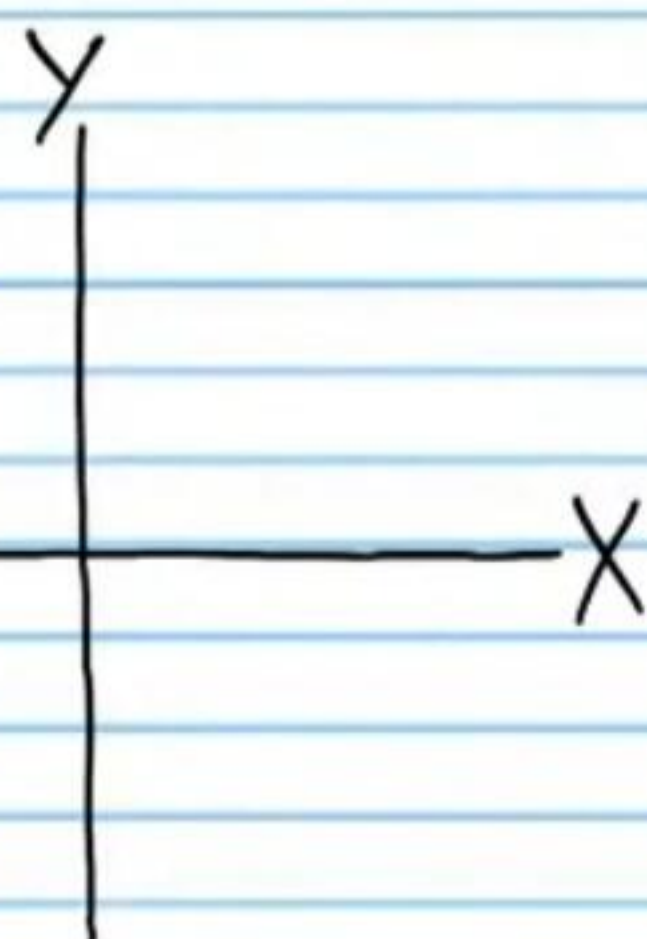
22) $\cos\left(-\frac{3\pi}{4}\right)$



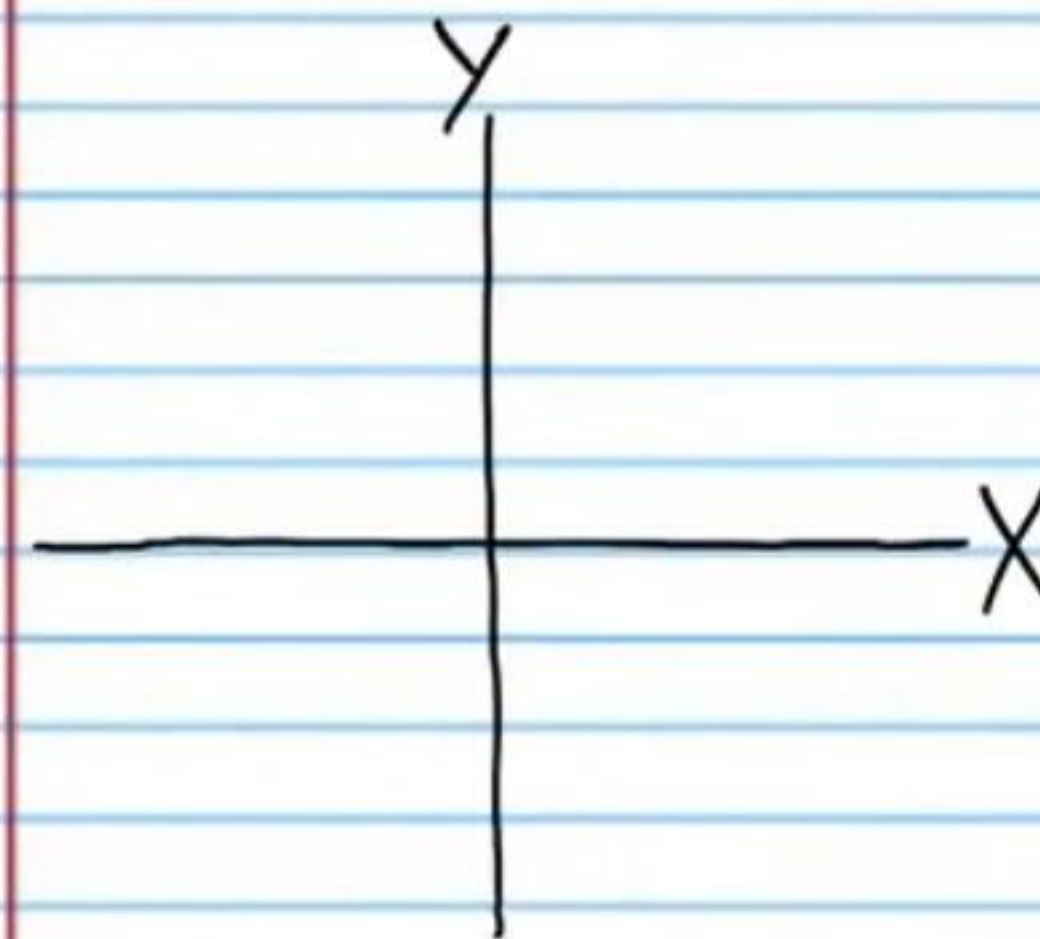
23) $\csc\left(\frac{5\pi}{3}\right)$



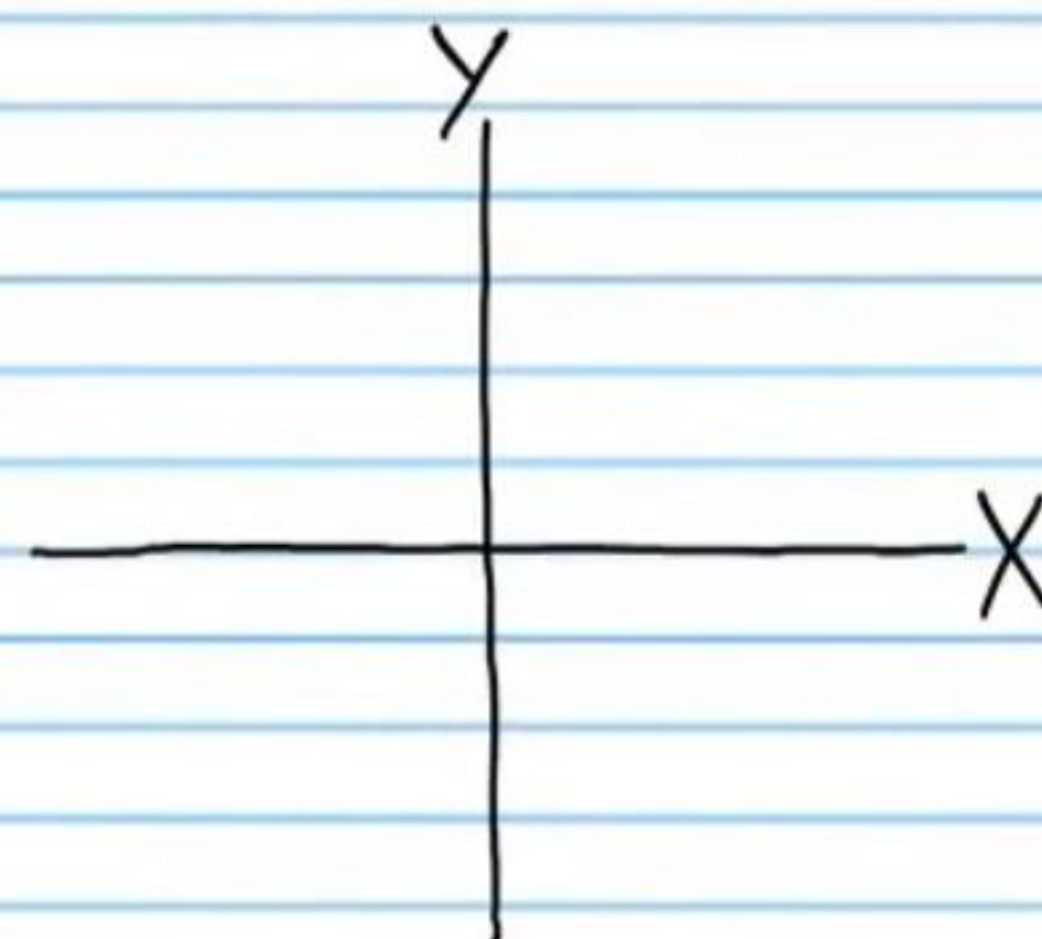
24) $\cot\left(\frac{\pi}{6}\right)$



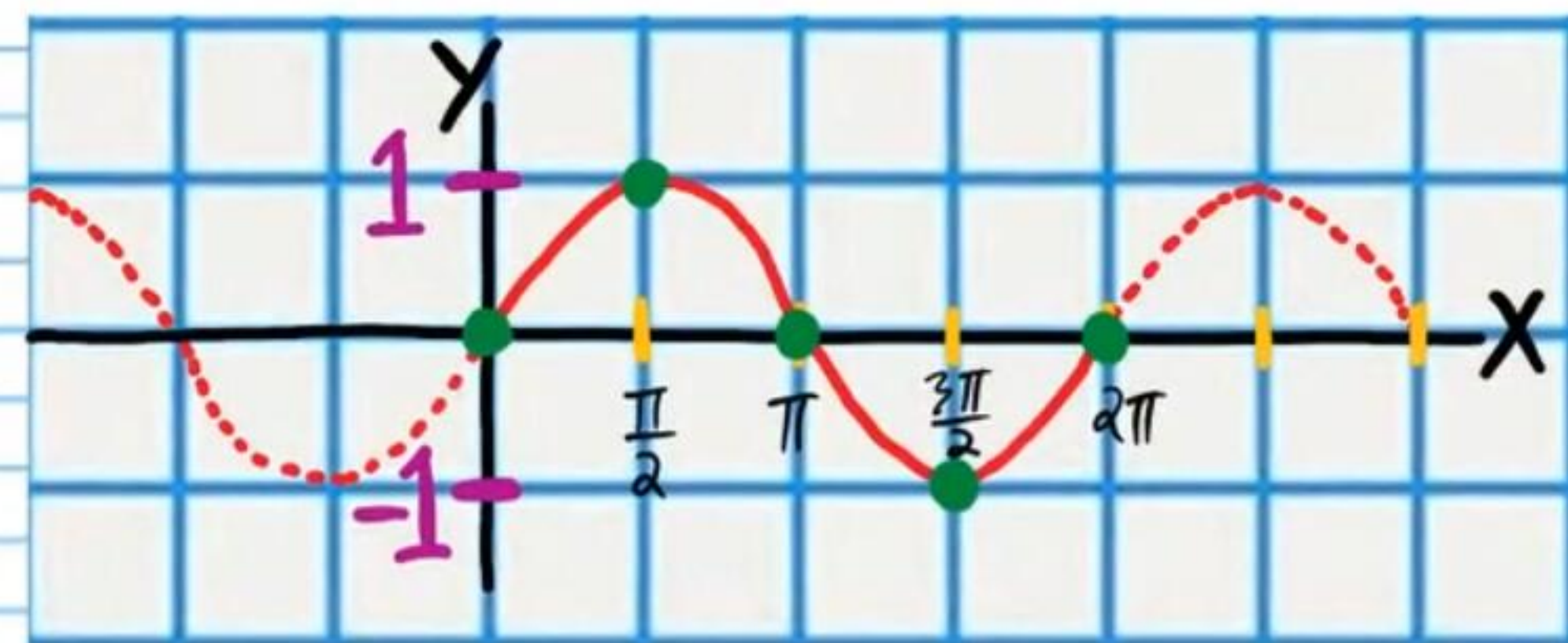
25) $\sin\left(\frac{7\pi}{4}\right)$



26) $\sec\left(-\frac{4\pi}{3}\right)$



$y = \sin x$



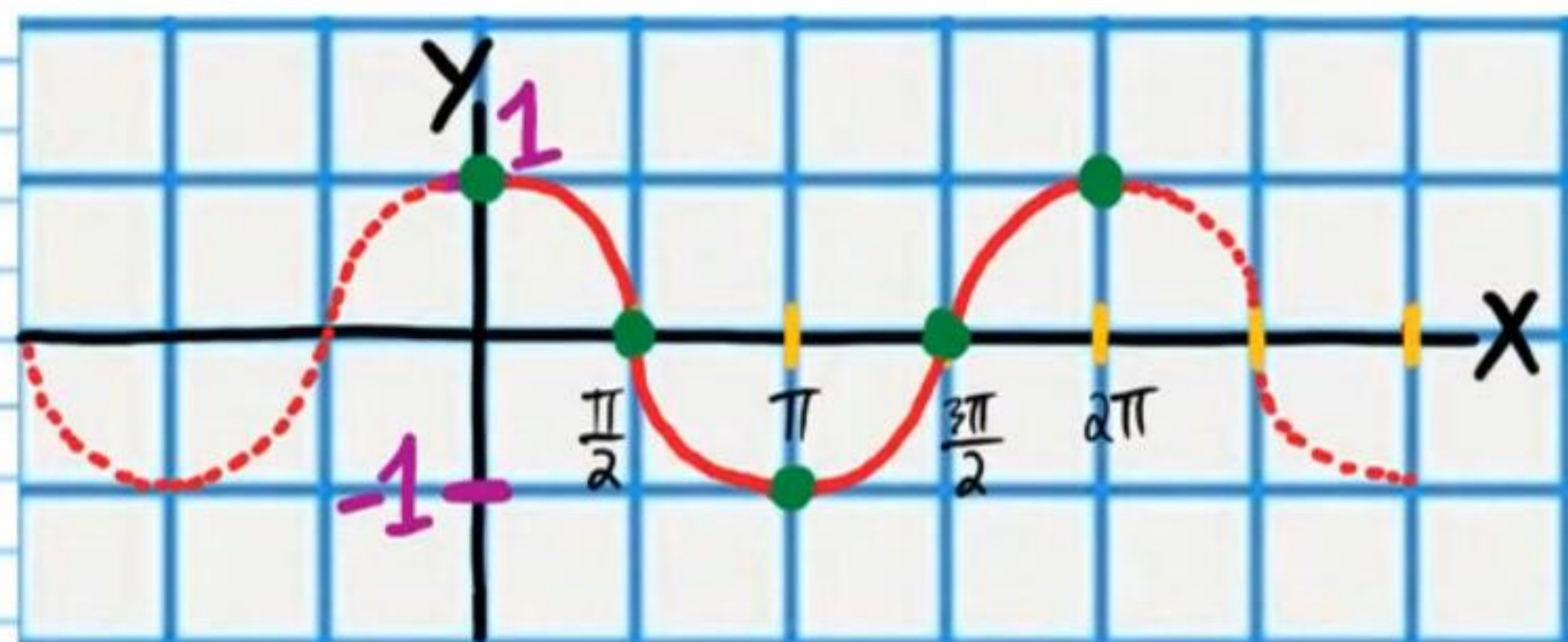
Use the graph above to find the following values.

27) $\sin\left(\frac{\pi}{2}\right)$

28) $\sin\left(-\frac{3\pi}{2}\right)$

29) $\sin \pi$

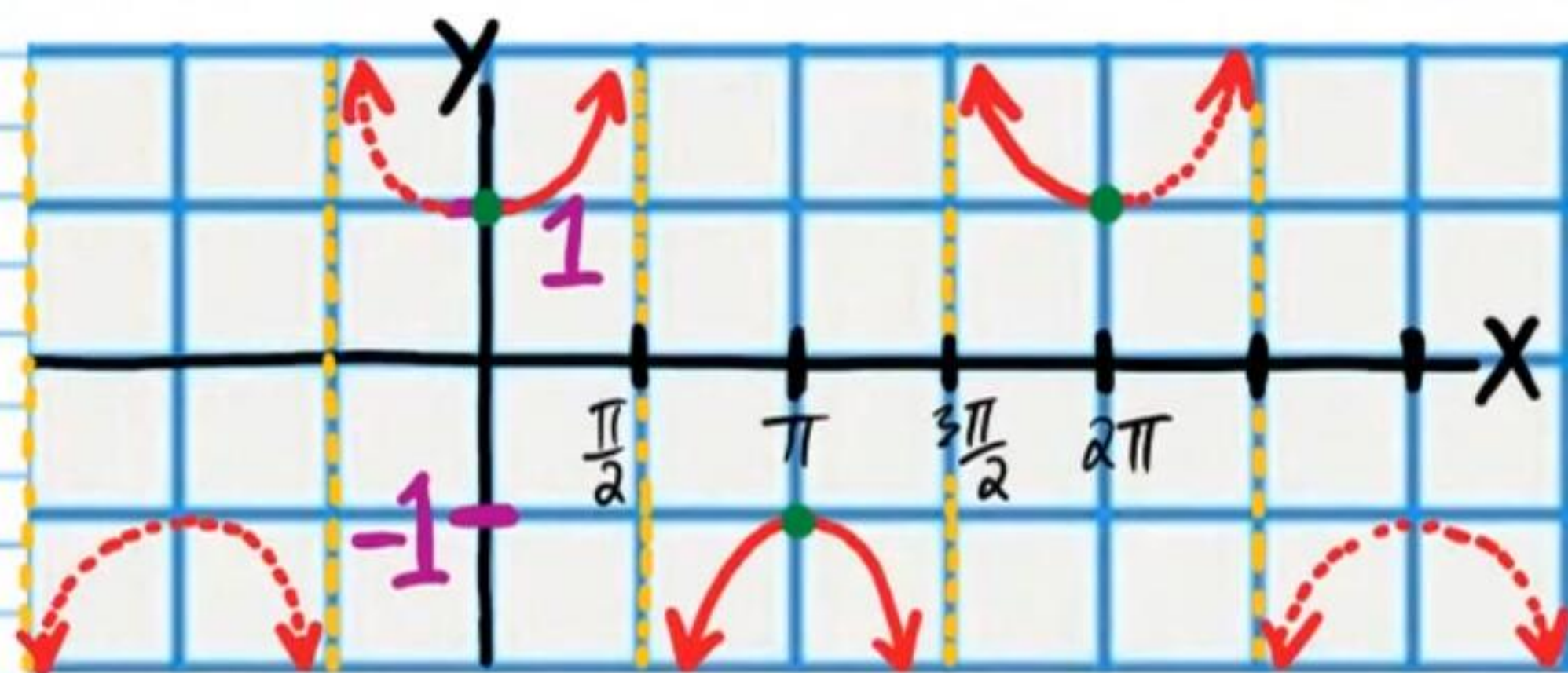
$$y = \cos x$$



Use the graph above to find the following values.

- (30) $\cos(-\pi)$ (31) $\cos(-\frac{3\pi}{2})$ (32) $\cos(\frac{5\pi}{2})$

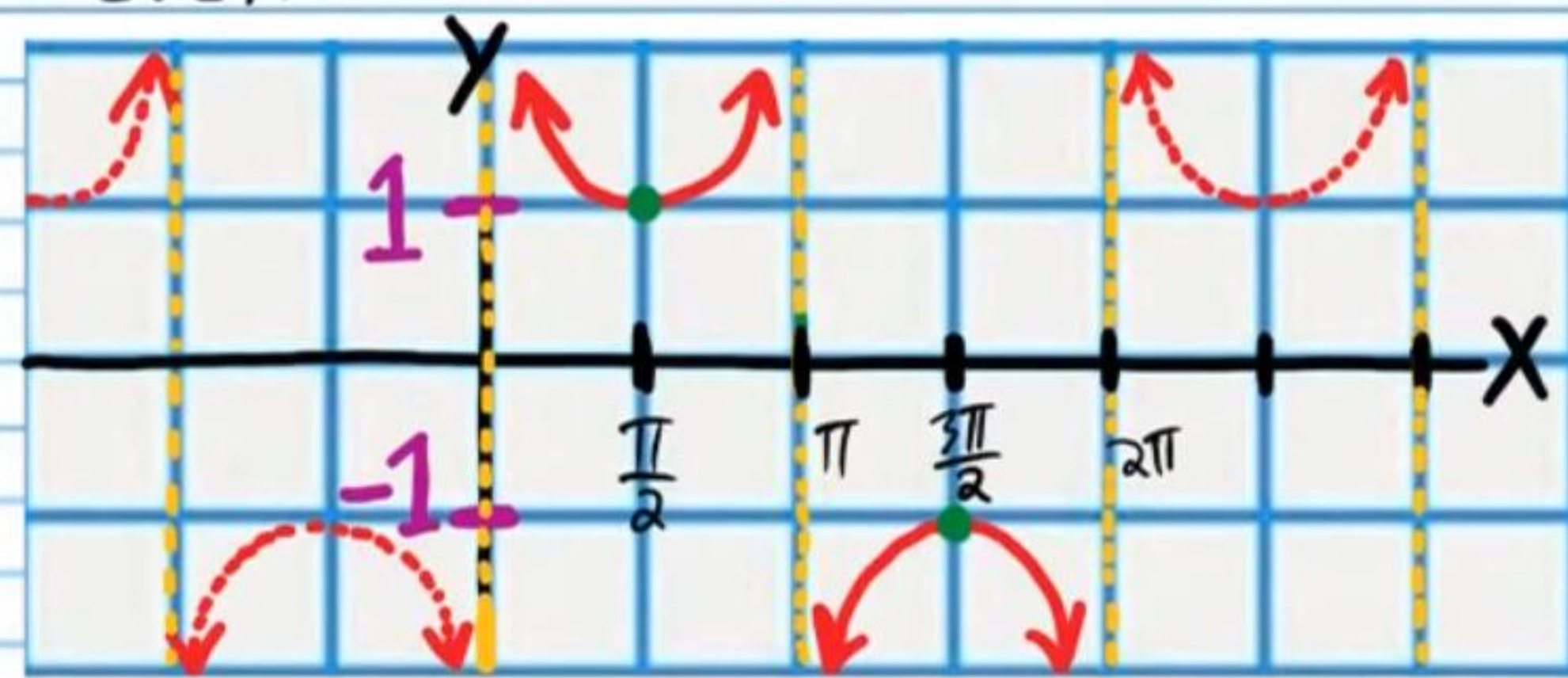
$$y = \sec x$$



Use the graph above to find the following values.

- (33) $\sec(\frac{5\pi}{2})$ (34) $\sec(2\pi)$ (35) $\sec(-\frac{3\pi}{2})$

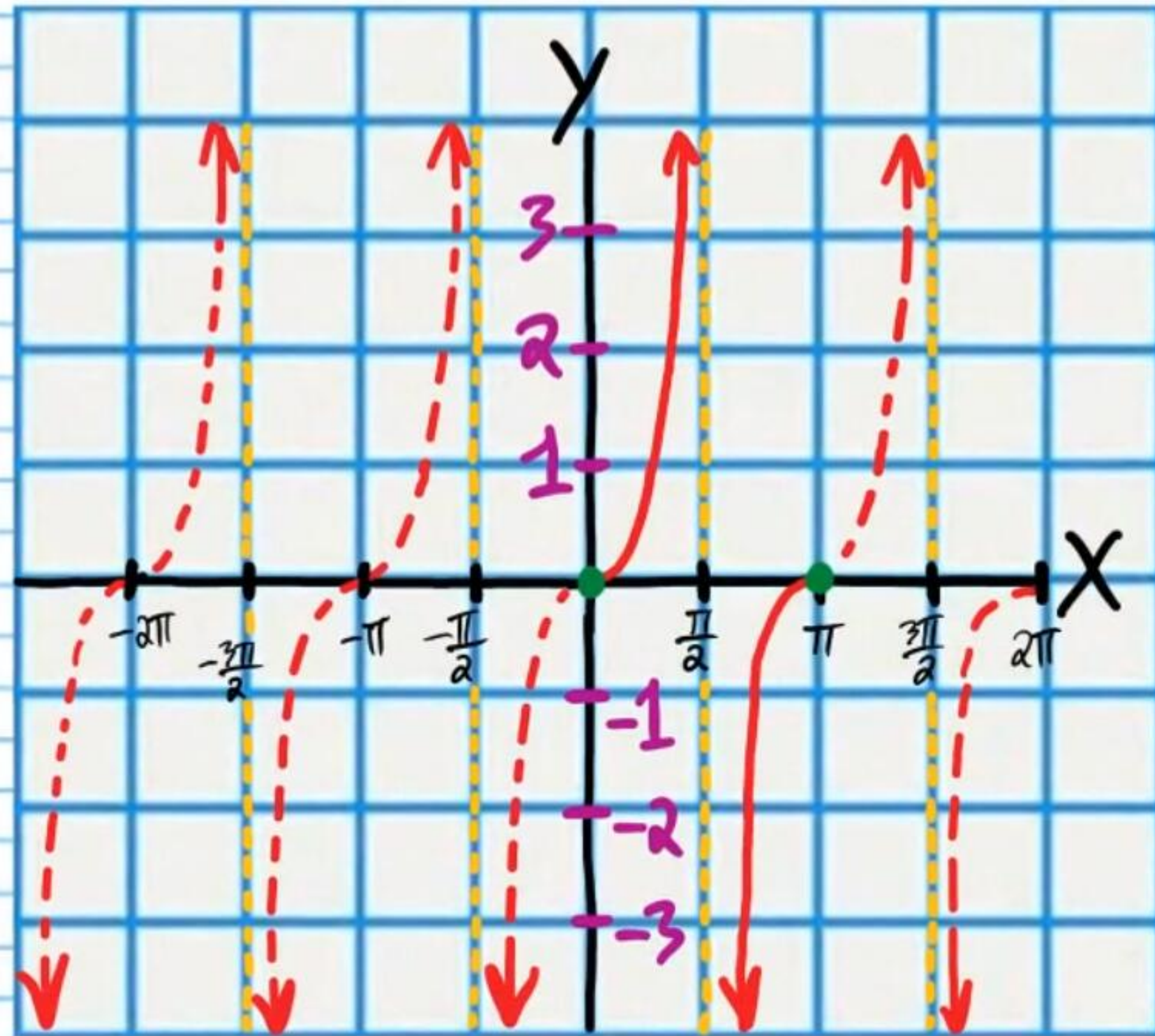
$$y = \csc x$$



Use the graph above to find the following values.

- (36) $\csc(-\pi)$ (37) $\csc(\frac{5\pi}{2})$ (38) $\csc \pi$

$$y = \tan x$$



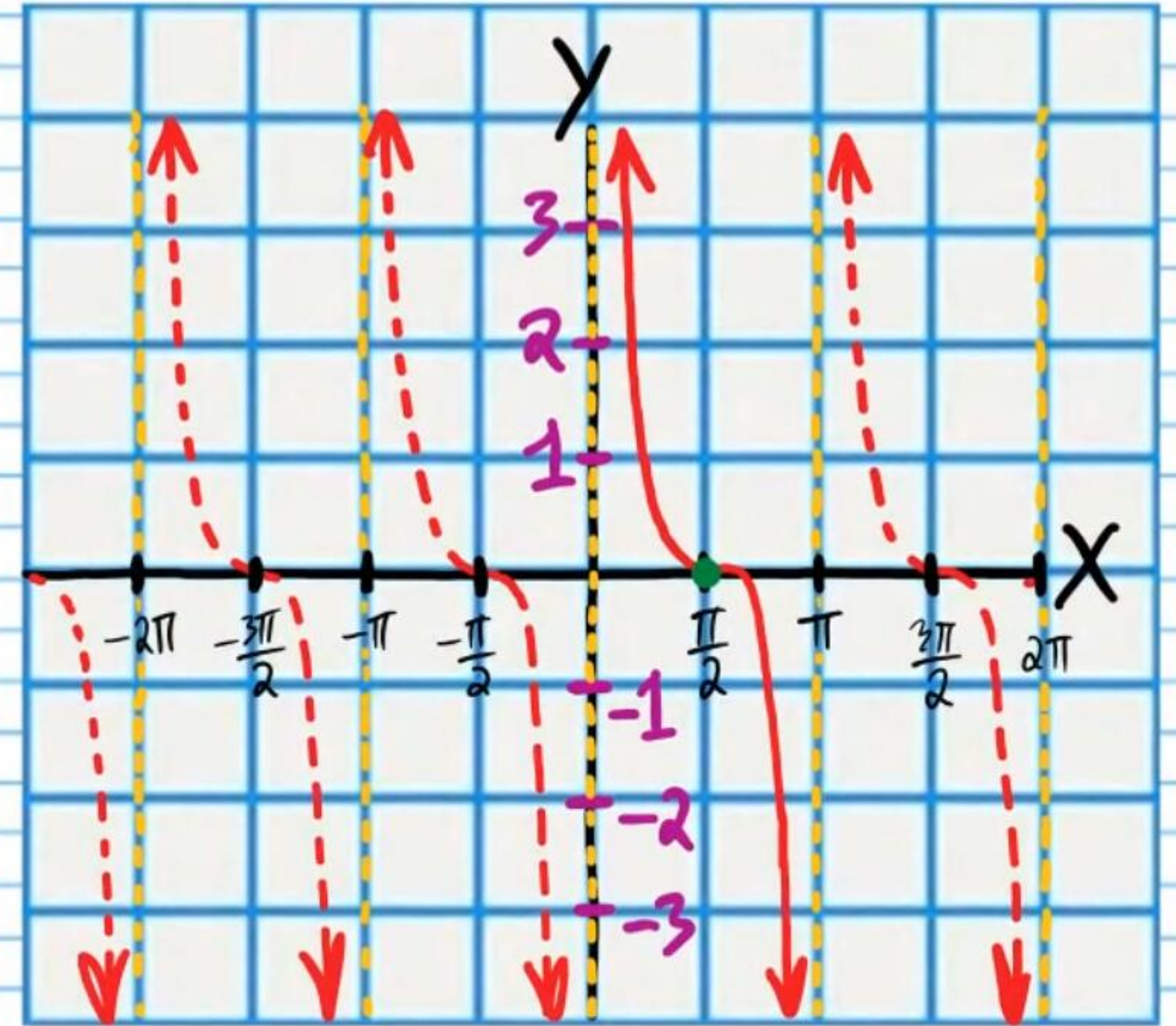
Use the graph above to find the following values.

③⑨ $\tan(0)$

④⑩ $\tan(-2\pi)$

④① $\tan\left(\frac{\pi}{2}\right)$

$$y = \cot x$$



Use the graph above to find the following values.

④② $\cot\left(\frac{\pi}{2}\right)$

④③ $\cot \pi$

④④ $\cot\left(-\frac{\pi}{2}\right)$

Answers

① $\frac{\sqrt[4]{x^{25}a^3}}{b^{12}}$

② $\frac{\sqrt{a^5} y^{16}}{\sqrt{b^{23}} x^{12}}$

GCF ③ $2x(4x^4 - 9x^3 - 50x^2 + 1)$

DOTS ④ $(b-7)(b+7)$

Quad ⑤ $(x+2)(x-1)$

DOTS ⑥ $(10x-9)(10x+9)$

Quad ⑦ $(x-2)(x-5)$

GCF ⑧ $5x^3(x^5 + 4x^3 + 6)$

Quad ⑨ $(3x+2)(x+2)$

Quad ⑩ $(2x-3)(5x+6)$

⑪ $\sqrt{70}$

⑫ $\sqrt[3]{2y^2}$

⑬ $\frac{3}{2}$

⑭ $4\sqrt{5}$

⑮ $3\sqrt[3]{6}$

⑯ $\frac{7\sqrt{5}}{5}$

⑰ $\frac{5\sqrt{10}}{20} = \frac{\sqrt{10}}{4}$

⑱ $\frac{3\sqrt{65}}{13}$

⑲ $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

⑳ -2

㉑ $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

㉒ $-\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$

㉓ $-\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$

㉔ $\sqrt{3}$

㉕ $-\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$

㉖ -2

㉗ 1

㉘ 1

㉙ 0

㉚ -1

㉛ 0

㉜ 0

㉝ und

㉞ 1

㉟ und

㊱ und

㊲ 1

㊳ und

㊴ 0

㊵ 0

㊶ und

㊷ 0

㊸ und

㊹ 0